

TOAMASINA, Madagascar

WMO: 670950

Lat: 18.110S

Lon: 49.393E

Elev: 22

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	62.0	62.9	59.0	74.5	67.1	60.5	78.7	67.2	21.1	73.3	18.9	73.1	1.9	240	0.280

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.2	89.3	79.8	88.0	79.6	87.5	79.4	81.6	86.5	80.8	86.0	80.1	85.4	10.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.3	157.5	85.6	79.1	151.4	84.5	78.8	149.6	84.1	45.4	86.5	44.6	86.8	43.9	85.5	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.5	16.0	14.1	59.2	94.8	1.1	3.5	58.4	97.4	57.7	99.4	57.1	101.4	56.3	104.0		
(5)			58.0	83.6	1.3	1.1	57.1	84.5	56.3	85.1	55.5	85.8	54.6	86.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.2	80.8	80.9	80.3	78.5	75.7	72.7	70.7	71.1	72.2	74.6	77.1	79.8		
	DBStd	4.16	1.66	1.84	1.65	1.78	2.13	2.11	1.83	1.73	1.74	2.17	2.07	1.66		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	9552	954	864	940	855	795	680	643	655	666	764	813	923		
	CDD65	4077	489	444	475	405	330	230	178	190	216	299	363	458		
	CDH74	31284	4939	4473	4462	3154	1906	868	486	638	959	1924	2983	4492		
	CDH80	8191	1655	1538	1399	857	350	62	7	8	27	216	646	1426		
(14) Wind	WSAvg	5.3	5.1	5.4	5.1	5.0	5.2	5.8	6.5	5.5	4.9	5.0	4.8	4.8		
(15) Precipitation	PrecAvg	127.4	16.4	14.3	18.1	11.6	10.5	10.2	11.2	7.9	4.8	4.5	6.3	11.8		
	PrecMax	175.4	40.0	34.1	47.0	38.5	58.1	21.0	21.7	17.0	11.3	15.6	16.5	35.8		
	PrecMin	91.6	3.1	2.1	4.3	2.6	2.7	3.2	3.8	3.0	1.3	0.8	0.6	0.7		
	PrecStd	22.4	8.1	8.1	8.8	6.5	8.1	4.1	4.3	3.2	2.6	3.3	3.9	7.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.0	91.1	89.8	88.0	86.1	82.8	80.8	80.8	82.1	84.5	86.5	89.5		
		MCWB	79.4	80.1	79.6	78.9	77.6	74.8	72.3	72.1	73.0	76.0	78.3	79.4		
	2%	DB	88.6	89.3	88.1	86.3	84.4	81.8	79.1	79.3	80.7	82.8	85.7	87.9		
		MCWB	79.8	80.0	79.6	78.3	76.5	73.9	71.0	71.0	72.2	75.0	77.8	79.5		
	5%	DB	87.8	88.0	87.4	85.8	83.2	80.6	78.4	78.8	79.8	82.3	84.4	86.6		
		MCWB	79.7	79.7	79.4	78.0	75.7	72.9	70.6	70.6	71.7	74.6	76.6	78.6		
	10%	DB	86.4	87.3	86.1	84.4	82.3	79.0	77.0	77.4	78.8	80.9	83.6	85.9		
		MCWB	79.0	79.4	78.6	77.1	75.1	71.8	70.0	69.8	71.2	73.4	75.9	78.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	82.6	82.6	82.0	80.7	79.2	76.9	74.5	74.1	74.9	78.2	80.1	81.7		
		MCDB	87.3	87.5	86.4	85.4	83.4	81.6	78.5	78.8	79.7	82.7	84.9	86.6		
	2%	WB	81.4	81.4	80.9	79.7	78.0	75.2	72.9	72.7	73.6	76.7	78.9	80.6		
		MCDB	86.4	86.7	85.9	84.7	82.7	79.9	77.4	77.8	78.8	81.4	83.7	86.0		
	5%	WB	80.6	80.6	80.1	78.9	76.8	74.1	71.8	71.7	72.7	75.7	77.9	79.8		
		MCDB	85.8	86.2	85.2	84.1	81.8	78.9	76.1	76.7	78.0	80.5	82.8	85.2		
	10%	WB	79.8	79.9	79.5	78.1	75.9	72.8	70.7	70.8	71.9	74.7	77.0	79.0		
		MCDB	84.9	85.3	84.5	83.2	80.8	77.5	75.2	75.8	77.2	79.6	81.9	84.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.2	12.1	12.3	12.6	12.5	12.7	13.4	13.6	13.2	12.9	12.5		
		MCDBR	13.7	13.7	13.5	13.4	14.2	14.4	14.9	14.8	14.9	13.7	13.5	13.3		
	5% WB	MCWBR	6.7	6.8	6.6	6.8	7.6	7.6	8.2	8.0	7.9	7.2	7.0	6.8		
		MCWBR	12.5	12.8	12.3	12.3	12.9	13.0	13.2	13.5	13.7	12.5	12.2	12.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.398	0.403	0.392	0.372	0.362	0.349	0.348	0.351	0.371	0.396	0.406	0.400		
		taud	2.549	2.536	2.568	2.614	2.625	2.625	2.614	2.580	2.503	2.446	2.445	2.509		
	Ebn at Noon	Ebn at Noon	300	296	292	286	278	276	280	289	294	294	296	299		
		Edh at Noon	35	35	33	29	27	26	27	30	34	38	38	36		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1861	1806	1655	1496	1236	1123	1143	1387	1682	1865	2014	2054		
	RadStd	RadStd	148	145	110	106	87	57	76	53	79	101	119	140		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.70	N/A	N/A	+0.59	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page